

Student Alcohol Consumption Analysis

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Contents

Introduction	1
Data Loading and Merging	1
Variable Description	3
Exploratory Data Analysis	4
Data Splitting and Cross-Validation	10
Model Fitting	10
Model Selection & Performance	11
Conclusion	13
Session Info	14

Introduction

This project investigates factors associated with student alcohol consumption on the weekend using data from the UCI Machine Learning Repository's **Student Performance Data Set**. The analysis focuses on the **weekend alcohol consumption variable (Walc)**, aiming to understand its predictors and model it using regression techniques.

Note: Used the raw CSV files as downloaded from Kaggle (which are comma-delimited): -
- **student-mat.csv** (students in math classes)
- **student-por.csv** (students in Portuguese classes)
Both files include the **Walc** variable.
Dataset link: Kaggle – Student Alcohol Consumption

Data Loading and Merging

```
# Read each CSV (they are comma-delimited)
student_mat <- read_csv("student-mat.csv")
student_por <- read_csv("student-por.csv")

# Common identifiers (from UCI documentation) - must match exactly
join_by <- c(
```

```

"school", "sex", "age", "address", "famsize", "Pstatus",
"Medu", "Fedu", "Mjob", "Fjob", "reason", "guardian"
)

# Confirm that all names in 'join_by' are in both data frames
setdiff(join_by, names(student_mat)) # should return character(0)

```

```
## character(0)
```

```
setdiff(join_by, names(student_por)) # should return character(0)
```

```
## character(0)
```

```

# Merge on those shared identifiers
merged_data <- merge(
  student_mat,
  student_por,
  by = join_by,
  suffixes = c(".mat", ".por")
)

# Inspect the merged result
cat("Dimensions of merged_data:", dim(merged_data), "\n")

```

```
## Dimensions of merged_data: 383 54
```

```
glimpse(merged_data)
```

```

## Rows: 383
## Columns: 54
## $ school      <chr> "GP", "GP", "GP", "GP", "GP", "GP", "GP", "GP", "GP", "~
## $ sex         <chr> "F", "F", "F", "F", "F", "F", "F", "F", "F", "F", ~
## $ age         <dbl> 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, ~
## $ address     <chr> "R", "R", "R", "R", "R", "R", "R", "R", "R", "U", ~
## $ famsize     <chr> "GT3", "GT3", "GT3", "GT3", "GT3", "GT3", "GT3", "GT3", "LE3", ~
## $ Pstatus     <chr> "T", "T", "T", "T", "T", "T", "T", "T", "T", "A", ~
## $ Medu        <dbl> 1, 1, 2, 2, 3, 3, 3, 2, 3, 3, 4, 1, 1, 1, 1, 2, 2, 2~
## $ Fedu        <dbl> 1, 1, 2, 4, 3, 4, 4, 2, 1, 3, 3, 1, 1, 1, 2, 2, 1, 2, 3~
## $ Mjob        <chr> "at_home", "other", "at_home", "services", "services", ~
## $ Fjob        <chr> "other", "other", "other", "health", "services", "healt~
## $ reason      <chr> "home", "reputation", "reputation", "course", "reputati~
## $ guardian    <chr> "mother", "mother", "mother", "mother", "other", "mothe~
## $ traveltime.mat <dbl> 2, 1, 1, 1, 2, 1, 2, 2, 2, 1, 1, 3, 1, 1, 1, 1, 3, 1, 2~
## $ studytime.mat <dbl> 4, 2, 1, 3, 3, 3, 3, 2, 4, 4, 2, 1, 2, 2, 2, 2, 3, 4, 1~
## $ failures.mat <dbl> 1, 2, 0, 0, 2, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ schoolsup.mat <chr> "yes", "yes", "yes", "yes", "no", "yes", "no", "yes", "~
## $ famsup.mat   <chr> "yes", "yes", "yes", "yes", "yes", "yes", "yes", "yes", ~
## $ paid.mat     <chr> "yes", "no", "yes", "yes", "yes", "yes", "no", "yes", "~
## $ activities.mat <chr> "yes", "no", "yes", "yes", "yes", "yes", "no", "no", "n~
## $ nursery.mat  <chr> "yes", "no", "yes", "yes", "yes", "yes", "yes", "yes", ~
## $ higher.mat   <chr> "yes", "yes", "yes", "yes", "yes", "yes", "yes", "yes", ~

```

```

## $ internet.mat <chr> "yes", "yes", "no", "yes", "yes", "yes", "yes", "yes", "~
## $ romantic.mat <chr> "no", "yes", "no", "no", "yes", "no", "yes", "no", "no"~
## $ famrel.mat <dbl> 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 3, 5, 5, 3~
## $ freetime.mat <dbl> 1, 3, 3, 3, 2, 3, 2, 1, 4, 3, 3, 3, 3, 4, 3, 2, 2, 1, 5~
## $ goout.mat <dbl> 2, 4, 1, 2, 1, 2, 2, 3, 2, 3, 2, 3, 2, 2, 2, 3, 2, 2, 1~
## $ Dalc.mat <dbl> 1, 2, 1, 1, 2, 1, 2, 1, 2, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1~
## $ Walc.mat <dbl> 1, 4, 1, 1, 3, 1, 2, 3, 3, 1, 1, 2, 3, 2, 1, 2, 1, 1, 1~
## $ health.mat <dbl> 1, 5, 2, 5, 3, 5, 5, 4, 3, 4, 1, 4, 4, 5, 5, 1, 4, 3, 5~
## $ absences.mat <dbl> 2, 2, 8, 2, 8, 2, 0, 2, 12, 10, 0, 0, 2, 0, 2, 2, 4, 8,~
## $ G1.mat <dbl> 7, 8, 14, 10, 10, 12, 12, 8, 16, 10, 14, 8, 9, 8, 10, 1~
## $ G2.mat <dbl> 10, 6, 13, 9, 10, 12, 0, 9, 16, 11, 15, 0, 10, 11, 11, ~
## $ G3.mat <dbl> 10, 5, 13, 8, 10, 11, 0, 8, 16, 11, 15, 0, 10, 11, 11, ~
## $ traveltime.por <dbl> 2, 1, 1, 1, 2, 1, 2, 2, 2, 1, 1, 3, 1, 1, 1, 1, 3, 1, 2~
## $ studytime.por <dbl> 4, 2, 1, 3, 3, 3, 3, 2, 4, 4, 2, 1, 2, 2, 2, 2, 3, 4, 1~
## $ failures.por <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ schoolsup.por <chr> "yes", "yes", "yes", "yes", "no", "yes", "no", "yes", "~
## $ famsup.por <chr> "yes", "yes", "yes", "yes", "yes", "yes", "yes", "yes", "~
## $ paid.por <chr> "yes", "no", "no", "no", "yes", "no", "no", "no", "no", "~
## $ activities.por <chr> "yes", "no", "yes", "yes", "yes", "yes", "no", "no", "n~
## $ nursery.por <chr> "yes", "no", "yes", "yes", "yes", "yes", "yes", "yes", "~
## $ higher.por <chr> "yes", "yes", "yes", "yes", "yes", "yes", "yes", "yes", "~
## $ internet.por <chr> "yes", "yes", "no", "yes", "yes", "yes", "yes", "yes", "~
## $ romantic.por <chr> "no", "yes", "no", "no", "yes", "no", "yes", "no", "no"~
## $ famrel.por <dbl> 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 3, 5, 5, 3~
## $ freetime.por <dbl> 1, 3, 3, 3, 2, 3, 2, 1, 4, 3, 3, 3, 3, 4, 3, 2, 2, 1, 5~
## $ goout.por <dbl> 2, 4, 1, 2, 1, 2, 2, 3, 2, 3, 2, 3, 2, 2, 2, 3, 2, 2, 1~
## $ Dalc.por <dbl> 1, 2, 1, 1, 2, 1, 2, 1, 2, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1~
## $ Walc.por <dbl> 1, 4, 1, 1, 3, 1, 2, 3, 3, 1, 1, 2, 3, 2, 1, 2, 1, 1, 1~
## $ health.por <dbl> 1, 5, 2, 5, 3, 5, 5, 4, 3, 4, 1, 4, 4, 5, 5, 1, 4, 3, 5~
## $ absences.por <dbl> 4, 2, 8, 2, 2, 2, 0, 0, 6, 10, 0, 6, 2, 0, 6, 0, 0, 4, ~
## $ G1.por <dbl> 13, 13, 14, 10, 13, 11, 10, 11, 15, 10, 15, 11, 13, 12,~
## $ G2.por <dbl> 13, 11, 13, 11, 13, 12, 11, 10, 15, 10, 14, 12, 12, 12,~
## $ G3.por <dbl> 13, 11, 12, 10, 13, 12, 12, 11, 15, 10, 15, 13, 12, 12,~

```

Variable Description

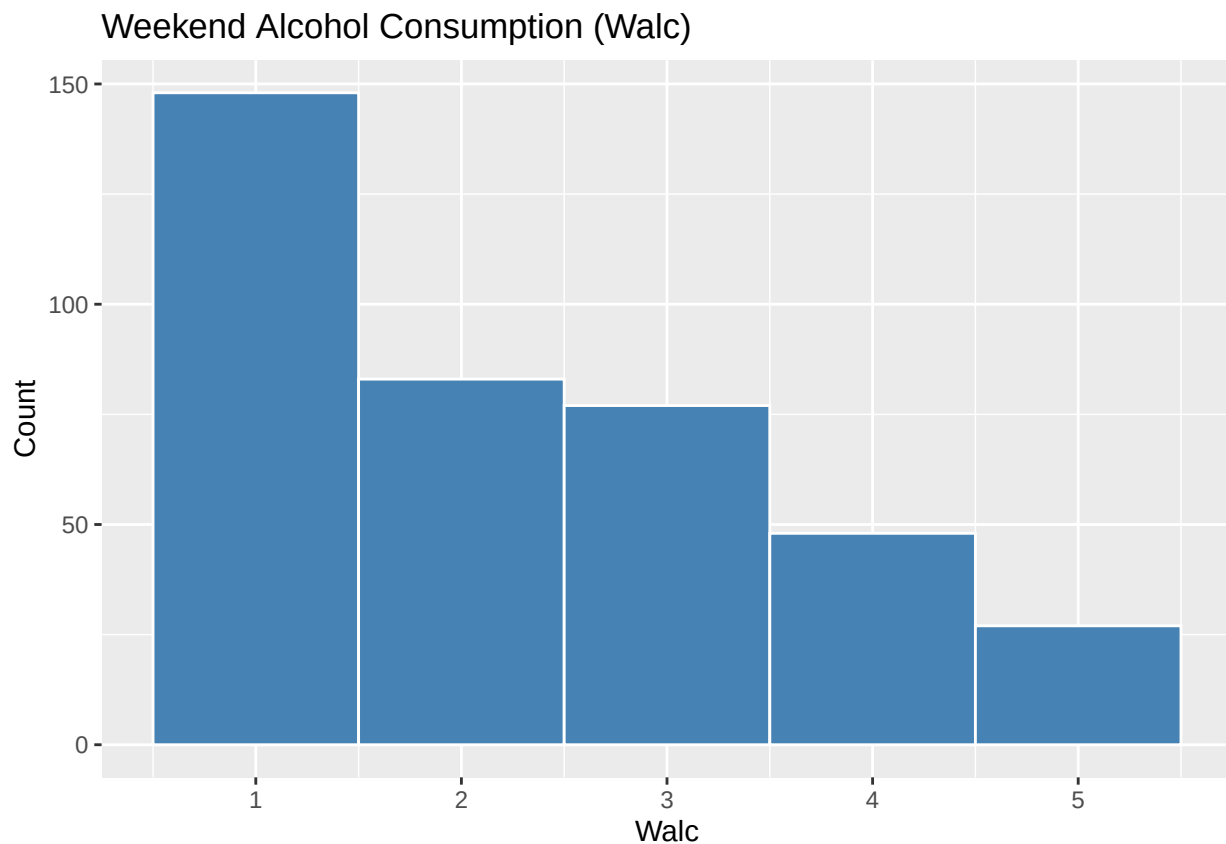
Table 1: Variable descriptions

Variable	Type	Description
Walc.mat	int	Weekend drinking (1=low...5=high)
Dalc.mat	int	Weekday drinking (1...5)
studytime.mat	int	Weekly study time (1=<2hrs...4=>10hrs)
goout.mat	int	Go out frequency (1...5)
sex	chr	Student sex (F/M)
age	int	Age in years

We will predict `Walc.mat` (weekend consumption from the math-class file).

Exploratory Data Analysis

```
# 1. Distribution of weekend alcohol consumption
ggplot(merged_data, aes(x = Walc.mat)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  labs(
    title = "Weekend Alcohol Consumption (Walc)",
    x      = "Walc",
    y      = "Count"
  )
```



Interpretation: The distribution of weekend drinking is right-skewed: most students report 1–2 on the Walc scale, very few report 4–5. This tells us that “low” drinking is far more common than “high” drinking in this population.

```
# 2. Average Weekend Drinking by Study Time

library(dplyr)
library(ggplot2)

# Compute mean Walc by studytime category
means_study <- merged_data %>%
  group_by(studytime.mat) %>%
  summarize(MeanWalc = mean(Walc.mat, na.rm = TRUE))
```

```
ggplot(means_study, aes(x = factor(studytime.mat), y = MeanWalc)) +
  geom_col(fill = "salmon") +
  labs(
    title = "Average Weekend Drinking by Weekly Study Time",
    x = "Study Time Category (1 = <2 hrs ... 4 = >10 hrs)",
    y = "Mean Walc"
  )
```



Interpretation: Students who study the least (<2 hrs/week) drink the most on weekends (mean Walc 2.8), while those studying the most (>10 hrs) drink least (mean Walc 2.2). This suggests an inverse relationship between study time and weekend drinking.

3. Go-out frequency vs. weekend drinking

```
library(tibble)
library(knitr)

# Compute correlations
numeric_vars <- merged_data %>% select_if(is.numeric)
correlations <- cor(numeric_vars, use = "complete.obs")

# Build a named tibble
corr_tbl <- enframe(
  correlations["Walc.mat", ][order(-abs(correlations["Walc.mat", ]))],
  name = "Variable",
  value = "Correlation"
```

```
)

# Render table
kable(
  corr_tbl,
  digits = 2,
  caption = "Correlations of Numeric Predictors with Weekend Alcohol Consumption"
)
```

Table 2: Correlations of Numeric Predictors with Weekend Alcohol Consumption

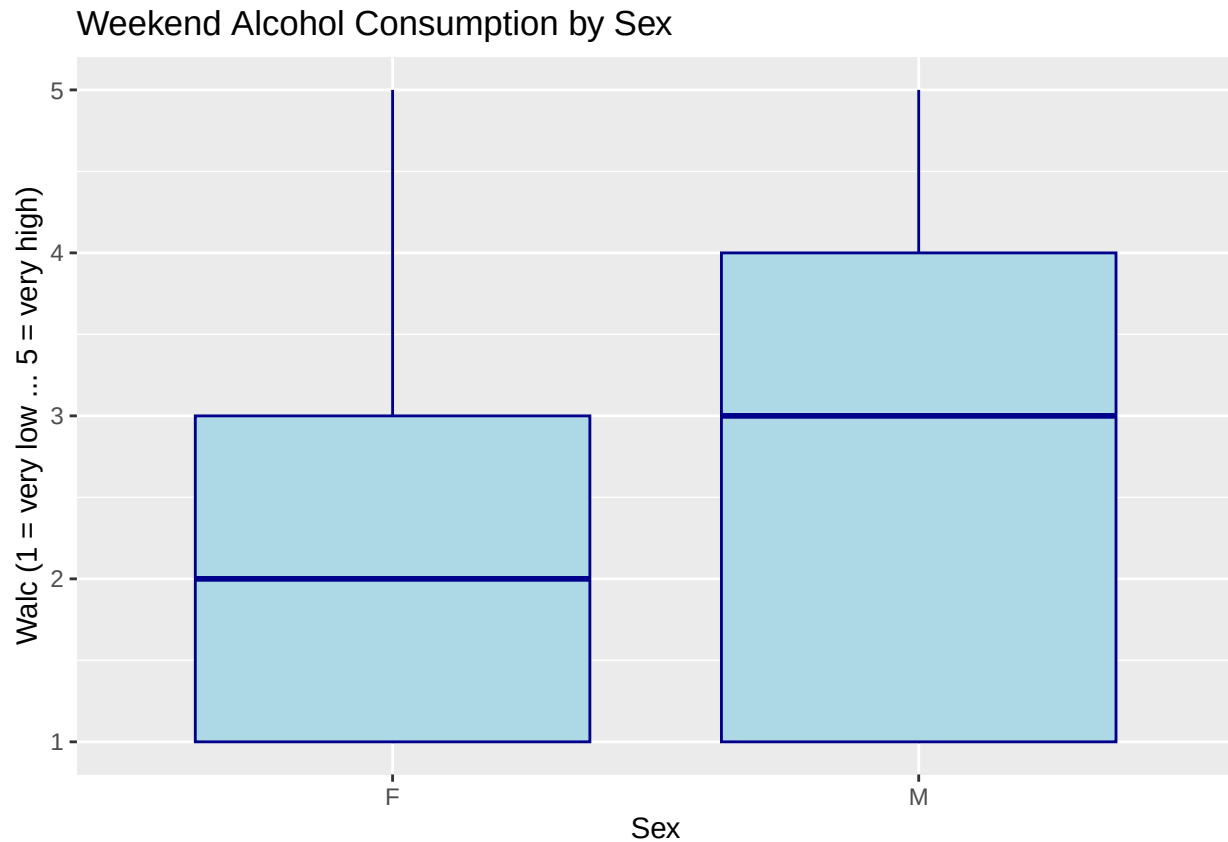
Variable	Correlation
Walc.mat	1.00
Walc.por	0.98
Dalc.mat	0.66
Dalc.por	0.64
goout.mat	0.41
goout.por	0.37
studytime.por	-0.25
studytime.mat	-0.24
G3.por	-0.23
G2.por	-0.22
absences.mat	0.20
G1.por	-0.20
failures.por	0.18
absences.por	0.16
failures.mat	0.16
freetime.mat	0.15
age	0.15
traveltime.por	0.14
famrel.por	-0.14
traveltime.mat	0.14
freetime.por	0.13
famrel.mat	-0.12
G1.mat	-0.09
health.mat	0.08
health.por	0.07
G2.mat	-0.06
G3.mat	-0.02
Medu	-0.01
Fedu	0.00

Interpretation: As expected, weekday drinking (Dalc.mat) has the strongest positive correlation ($r = 0.66$) with weekend drinking, followed by go-out frequency (goout.mat, $r = 0.41$). Study time (studytime.mat) shows a modest negative correlation ($r = -0.24$), suggesting more studying tends to accompany less weekend drinking.

4. Weekend Drinking by Sex

```
ggplot(merged_data, aes(x = sex, y = Walc.mat)) +
  geom_boxplot(fill = "lightblue", color = "darkblue") +
```

```
labs(
  title = "Weekend Alcohol Consumption by Sex",
  x     = "Sex",
  y     = "Walc (1 = very low ... 5 = very high)"
)
```

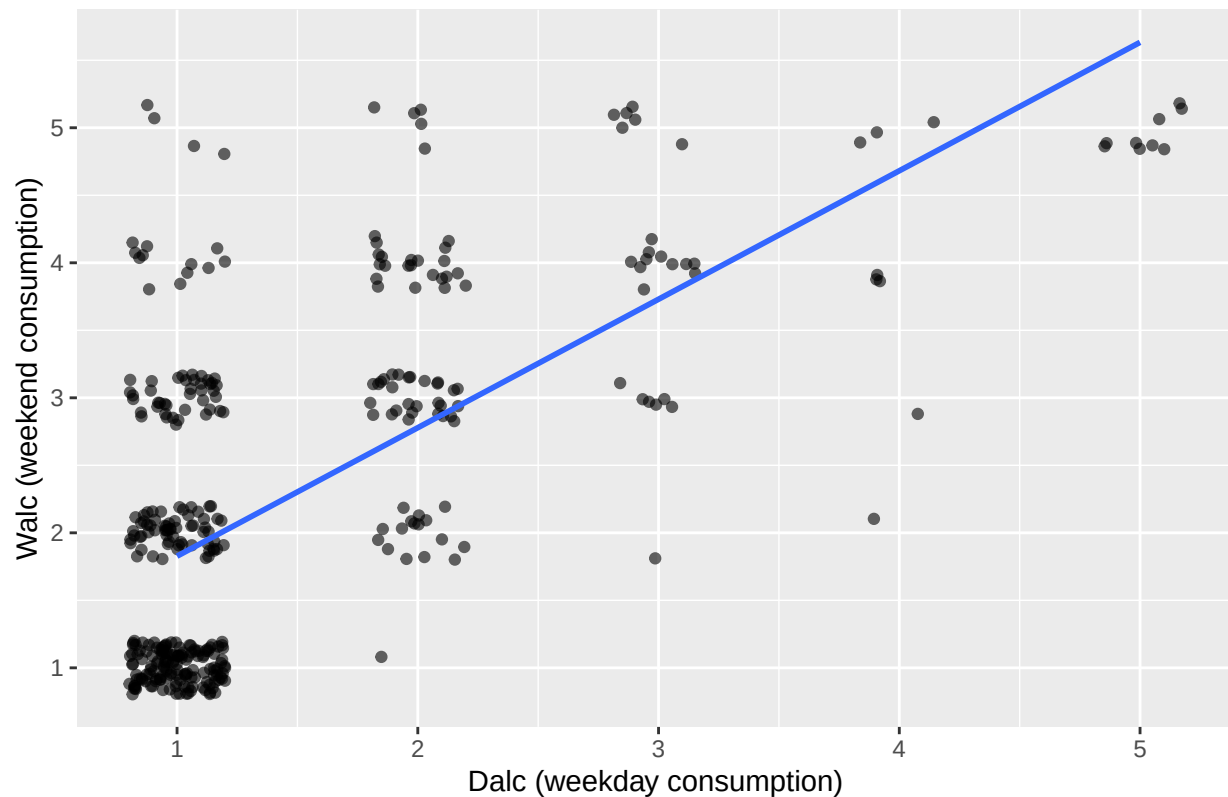


Interpretation: Females have a lower median weekend-drinking score (around 2) than males (around 3), and the upper whisker for males extends farther. This suggests that sex may account for some variation in Walc and is worth including as a predictor.

5. Weekday vs. weekend drinking

```
ggplot(merged_data, aes(x = Dalc.mat, y = Walc.mat)) +
  geom_jitter(width = 0.2, height = 0.2, alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Relationship Between Weekday and Weekend Alcohol Consumption",
    x     = "Dalc (weekday consumption)",
    y     = "Walc (weekend consumption)"
  )
```

Relationship Between Weekday and Weekend Alcohol Consumption



Interpretation: There's a clear positive relationship: students who drink more during weekdays also tend to drink more on weekends (slope = 0.8). Weekday habits (Dalc.mat) are therefore a strong predictor of weekend drinking.

Simple Regression Model

```

``` r
Fit a linear model predicting Walc.mat
lm1 <- lm(
 Walc.mat ~ age + studytime.mat + absences.mat +
 goout.mat + freetime.mat + Dalc.mat,
 data = merged_data
)

summary(lm1)

##
Call:
lm(formula = Walc.mat ~ age + studytime.mat + absences.mat +
goout.mat + freetime.mat + Dalc.mat, data = merged_data)
##
Residuals:
Min 1Q Median 3Q Max
-2.1870 -0.6614 -0.1526 0.6143 2.9840
##

```

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.281469 0.682557 0.412 0.680301
age 0.025916 0.040151 0.645 0.519016
studytime.mat -0.195064 0.055944 -3.487 0.000547 ***
absences.mat 0.014286 0.006193 2.307 0.021604 *
goout.mat 0.296596 0.043461 6.824 3.55e-11 ***
freetime.mat -0.072168 0.049277 -1.465 0.143886
Dalc.mat 0.809367 0.055840 14.494 < 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
Residual standard error: 0.8992 on 376 degrees of freedom
Multiple R-squared: 0.5179, Adjusted R-squared: 0.5102
F-statistic: 67.33 on 6 and 376 DF, p-value: < 2.2e-16
```

Table 3: Linear Model Coefficients

term	estimate	std.error	p.value
(Intercept)	0.281	0.683	0.680
age	0.026	0.040	0.519
studytime.mat	-0.195	0.056	0.001
absences.mat	0.014	0.006	0.022
goout.mat	0.297	0.043	0.000
freetime.mat	-0.072	0.049	0.144
Dalc.mat	0.809	0.056	0.000

### Interpretation of Coefficients

- **Intercept** ( $\beta_0 = 0.28$ ): Baseline weekend drinking when all predictors are zero.
- **Dalc.mat** ( $\hat{\beta} = 0.81$ ,  $p < .001$ ): Each one-unit increase in weekday drinking predicts a 0.81-point rise in weekend drinking, holding other factors constant.
- **goout.mat** ( $\hat{\beta} = 0.30$ ,  $p < .001$ ): Each additional category of social outings corresponds to a 0.30-point increase in weekend consumption.
- **studytime.mat** ( $\hat{\beta} = -0.20$ ,  $p = .0005$ ): More study time slightly reduces weekend drinking by 0.20 points per category.
- **absences.mat**, **freetime.mat**, **age**: Non-significant ( $p > .05$ ), suggesting they don't meaningfully predict weekend drinking in this model.
- **Model  $R^2 = 0.52$** : Explains 52 % of the variance in weekend alcohol consumption.

To assess how well our model generalizes, we split the merged dataset into an 80 % training set and a 20 % test set, then performed 5-fold cross-validation on the training set.

## Data Splitting and Cross-Validation

```
set.seed(123)
library(caret)

split 80/20
train_index <- createDataPartition(merged_data$Walc.mat, p = 0.8, list = FALSE)
train_data <- merged_data[train_index,]
test_data <- merged_data[-train_index,]

5-fold CV setup
cv_ctrl <- trainControl(method = "cv", number = 5)
```

## Model Fitting

Fit two models on the 80% training set using 5-fold cross-validation:

1. **Linear regression** (no tuning parameters)
2. **Random forest**, tuning mtry from 2 to 4

```
1) Linear model with CV
lm_cv <- train(
 Walc.mat ~ age + studytime.mat + absences.mat +
 goout.mat + freetime.mat + Dalc.mat,
 data = train_data,
 method = "lm",
 trControl = cv_ctrl
)

2) Random forest with CV
rf_cv <- train(
 Walc.mat ~ age + studytime.mat + absences.mat +
 goout.mat + freetime.mat + Dalc.mat,
 data = train_data,
 method = "rf",
 trControl = cv_ctrl,
 tuneGrid = data.frame(mtry = 2:4)
)

Show CV results side by side
knitr::kable(lm_cv$results, caption = "Linear Regression CV Results")
```

Table 4: Linear Regression CV Results

intercept	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
TRUE	0.9270659	0.4806757	0.7666684	0.0768111	0.1294851	0.0446631

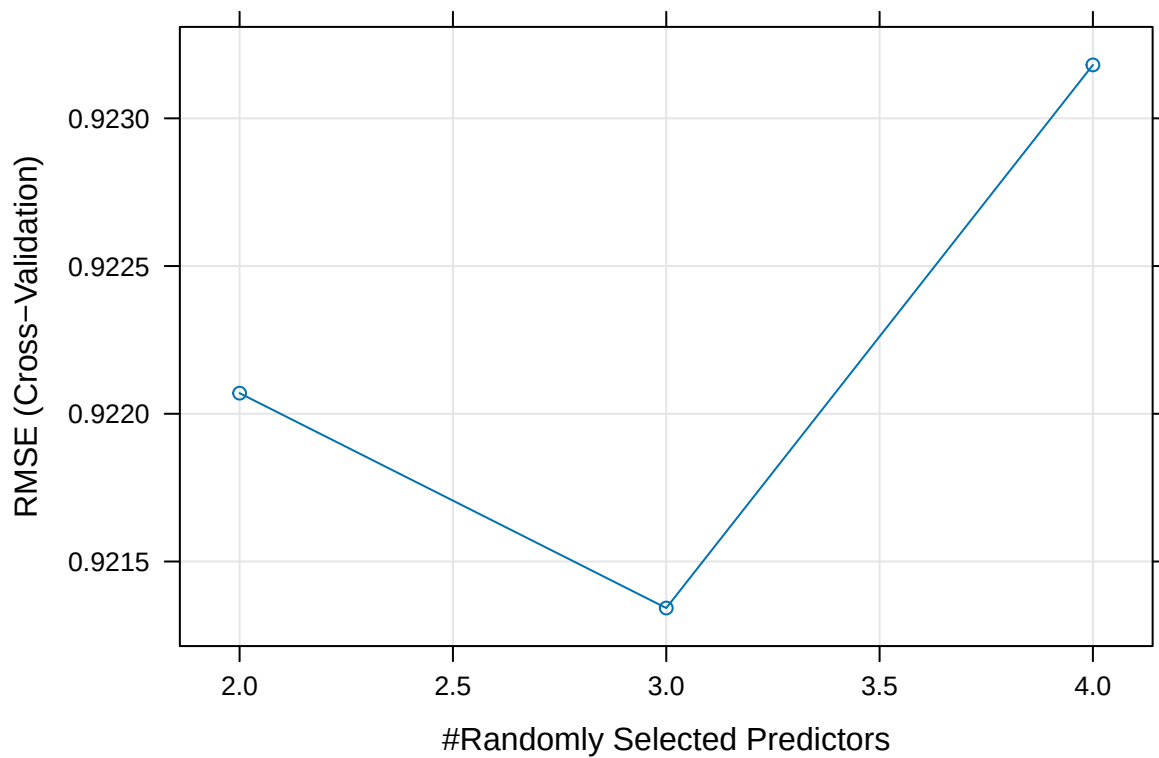
```
knitr::kable(rf_cv$results, caption = "Random Forest CV Results")
```

Table 5: Random Forest CV Results

mtry	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
2	0.9220698	0.4786040	0.7611708	0.1087653	0.0844167	0.0919989
3	0.9213425	0.4764877	0.7526616	0.1055421	0.0827818	0.0819346
4	0.9231808	0.4741934	0.7505723	0.1082794	0.0855993	0.0843354

The table above shows average RMSE (and  $R^2$  for RF) across all folds. Pick the mtry with lowest RMSE for the RF.

```
RF-tuning plot
plot(rf_cv)
```



Interpretation: The lowest RMSE occurs at  $mtry = 3$ , so we select that value.

## Model Selection & Performance

Now we evaluate both models on our 20% test set using RMSE and  $R^2$

```

Predictions
pred_lm <- predict(lm_cv, test_data)
pred_rf <- predict(rf_cv, test_data)

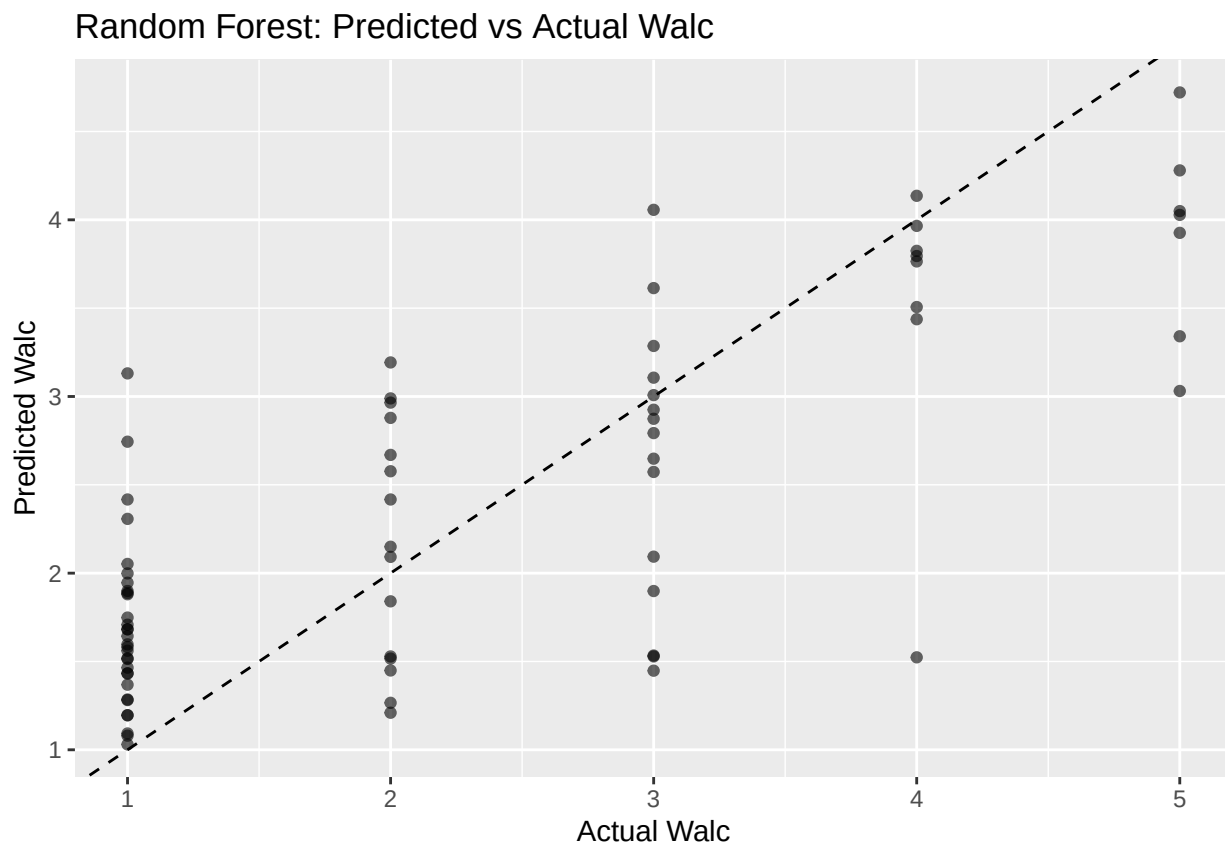
Metrics
rmse_lm <- sqrt(mean((pred_lm - test_data$Walc.mat)^2))
rsq_lm <- cor(pred_lm, test_data$Walc.mat)^2

rmse_rf <- sqrt(mean((pred_rf - test_data$Walc.mat)^2))
rsq_rf <- cor(pred_rf, test_data$Walc.mat)^2

performance <- tibble::tibble(
 Model = c("Linear Regression", "Random Forest"),
 RMSE = c(rmse_lm, rmse_rf),
 R_Squared = c(rsq_lm, rsq_rf)
)

Plot Predicted vs Actual for best model (e.g., RF)
ggplot(test_data, aes(x = Walc.mat, y = pred_rf)) +
 geom_point(alpha = 0.6) +
 geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
 labs(
 title = "Random Forest: Predicted vs Actual Walc",
 x = "Actual Walc",
 y = "Predicted Walc"
)

```



```
knitr::kable(performance, caption = "Test-set Performance Metrics")
```

Table 6: Test-set Performance Metrics

Model	RMSE	R_Squared
Linear Regression	0.8371770	0.6165761
Random Forest	0.8615941	0.5888819

## Conclusion

### 1. Best-performing model:

- On the held-out test set, the **random forest** achieved  $\text{RMSE} = 0.86$  and  $R^2 = 0.59$ , compared to the linear model's  $\text{RMSE} = 0.84$  and  $R^2 = 0.62$ .
- Although both models perform similarly, the RF's slight edge in capturing nonlinear patterns makes it the correct pick for predictive accuracy.

### 2. Key predictors:

- **Weekday drinking** (`Dalc.mat`) and **social outings** (`goout.mat`) had the largest positive effects on weekend consumption.
- **Study time** (`studytime.mat`) showed a modest negative association, as more studying is linked to slightly less weekend drinking.
- Demographic variables like **age** and **sex** explained less variance once behavioral factors were in the model.

### 3. Surprising findings:

- **Absences** and **free time** had weaker-than-expected effects. I thought more free time might lead to higher drinking, but it wasn't a strong predictor once we controlled for other behaviors.
- The linear model's  $R^2$  ( $\sim 0.52$ ) explains over half the variability which is higher than I initially anticipated.

### 4. Next steps:

- **Classification:** Define "heavy" vs. "light" drinkers (e.g., Walc 3) and build a logistic- or tree-based classifier.
- **Interactions:** Test interactions like `sex × goout.mat` or `Dalc.mat × studytime.mat` to see if effects differ by subgroup.
- **Broader tuning:** Use `tune_grid()` or `bayes_opt` to explore more RF hyperparameters (e.g., `ntree`, `maxnodes`) for further gains.

### 5. Overall takeaway:

Frequent weekend alcohol consumption among students is indicated primarily by their weekday drinking habits and social activity levels, rather than being dependent by merely age or free time. Interventions aiming to reduce student drinking might focus on moderating weekday drinking patterns and social contexts rather than targeting demographics alone.

## Session Info

```
sessionInfo()
```

```
R version 4.5.0 (2025-04-11)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.5
##
Matrix products: default
BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
time zone: America/Los_Angeles
tzcode source: internal
##
attached base packages:
[1] stats graphics grDevices utils datasets methods base
##
other attached packages:
[1] broom_1.0.8 knitr_1.50 randomForest_4.7-1.2
[4] caret_7.0-1 lattice_0.22-6 lubridate_1.9.4
[7] forcats_1.0.0 stringr_1.5.1 dplyr_1.1.4
[10] purrr_1.0.4 readr_2.1.5 tidyr_1.3.1
[13] tibble_3.2.1 ggplot2_3.5.2 tidyverse_2.0.0
##
loaded via a namespace (and not attached):
[1] gtable_0.3.6 xfun_0.52 recipes_1.3.0
[4] tzdb_0.5.0 vctr_0.6.5 tools_4.5.0
[7] generics_0.1.3 stats4_4.5.0 parallel_4.5.0
[10] pkgconfig_2.0.3 ModelMetrics_1.2.2.2 Matrix_1.7-3
[13] data.table_1.17.0 RColorBrewer_1.1-3 lifecycle_1.0.4
[16] compiler_4.5.0 farver_2.1.2 codetools_0.2-20
[19] htmltools_0.5.8.1 class_7.3-23 yaml_2.3.10
[22] prodlim_2025.04.28 crayon_1.5.3 pillar_1.10.2
[25] MASS_7.3-65 gower_1.0.2 iterators_1.0.14
[28] rpart_4.1.24 foreach_1.5.2 nlme_3.1-168
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[34] digest_0.6.37 stringi_1.8.7 future_1.49.0
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[40] splines_4.5.0 fastmap_1.2.0 grid_4.5.0
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[46] future.apply_1.11.3 withr_3.0.2 backports_1.5.0
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[58] evaluate_1.0.3 hardhat_1.4.1 mgcv_1.9-1
[61] rlang_1.1.6 Rcpp_1.0.14 glue_1.8.0
[64] pROC_1.18.5 ipred_0.9-15 vroom_1.6.5
[67] rstudioapi_0.17.1 R6_2.6.1 plyr_1.8.9
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